

Network Operations Strategy Memo — ETA Reliability & Bottleneck Action Plan

To: Head of Network Operations · **Basis:** 14,817 trips / 26,369 corridor legs, Sep 12 – Oct 3, 2018 · **Scope:** steady-state, non-peak (no festival-season data); all findings re-verified by full pipeline re-run

Three decisions, in priority order

1. **Stop quoting raw OSRM times — deploy the corridor calibration table this quarter.** OSRM misses the true transit time by 107 minutes per leg on average; only 4.5% of legs land within 15% of it. A simple table — each corridor’s historical actual-vs-OSRM multiplier — cuts that error to 36 minutes (47% within 15%) with no model, no infrastructure. The full prediction model adds only ~2 minutes more (54% within 15%): **the table is 95% of the win and can ship in weeks.**
2. **Quote the 80th-percentile time, not the average.** Today a median quote of 75 minutes is kept roughly half the time. Quoting 90 minutes (the p80 figure from the same data) is kept ~80% of the time — verified out-of-sample at 78%. The 15-minute longer quote is a pricing/positioning choice; the breach-rate drop is measured, not modeled.
3. **Fix three hubs and twelve city corridors — not a top-five list.** Three hubs are statistically solid bottlenecks (they survive 98–100% of resampling checks); ranks 4–6 are a tie and should be treated as one watch-tier. Legs touching the top three carry **39% of all network excess hours** (13,708 of 34,937 hours over 15 days) while being only 17.8% of legs.

The three hubs — and why they need different fixes

Hub	Excess hours /day touching it	Late legs /day	Median dwell	Diagnosis → intervention
Gurgaon Bilaspur HB	~566 (network #1)	18	2.5 h	Pass-through chokepoint (most through-traffic in network). Sortation/cross-dock capacity + parallel routing for trips that don’t terminate there
Bhiwandi Mankoli HB	~247	27 (network #1 in breaches)	2.4 h	Volume terminal, almost no through-traffic. Dock scheduling + unload capacity, not routing
Bangalore Nelmangala H	~287	9	2.4 h	Mixed profile. Dwell reduction first; ~2.4 h median dwell is the binding constraint

Dwell of ~2.4–2.5 hours per leg-start is near-uniform across all top hubs — handling time, not driving, is where the schedule dies.

The corridor problem is inside cities, not between them

Of 145 chronically delayed corridors (worst-decile delay ratios, min. 5 observations), **93% are intra-state and 68% are Carting** — short urban shuttle runs, not line-haul. The twelve worst breach their promise on **69–100% of runs** — e.g., Mumbai Hub→Mira Road late 31 of 45 runs, Hyderabad Chikkadpally→Shamshabad late 22 of 22. This is structural, so it is fixable: dispatch-window changes and load consolidation on ~12 named corridors, not a network-wide program.

Route-type rule: on comparable runs, FTL beats Carting’s delay ratio in 9 of 10 corridor profiles; distance-adjusted, Carting costs up to ~2 extra hours on long daytime corridors. At plausible costs the choice stays volume-driven (break-even $\approx$ FTL trip cost $\div$ Carting per-kg rate $\approx$ 2,000 kg at our dial settings) — **except long daytime corridors, where the delay penalty justifies FTL from about half-load.**

What the top-3 fix is worth (formula + dials, not invented precision)

Measured: 914 excess hours/day and 54 late legs/day (14% of all breaches) touch the top-3 hubs. Not in this dataset: shipment value, SLA penalties, recoverable share of handling time. Impact therefore comes as a formula with the dials exposed:

Quantity	Formula	Conservative (f=20%)	Ambitious (f=40%)
Hours recovered /day	$f \times 914$	183 h	366 h
Late deliveries avoided	$f \times 14\% \text{ of breaches}$	-2.8% network late rate	-5.6%
Revenue at risk recovered /day	$f \times 54 \text{ legs} \times \text{value} \times \text{at-risk}\%$	₹54k @ ₹10k/leg, 50% at-risk	₹108k

f = the share of hub-attributed excess an upgrade actually removes — set it from a 2-week dwell time-study, not from this memo. Excess time vs OSRM partly reflects unavoidable handling; claiming all 914 hours would be dishonest.

90-day plan

When	Action	Owner	Success measure
Wks 1–4	Calibration table live in quoting; p80 quotes A/B on 2 regions	Network planning	Within-15% rate 4.5% → >45%; kept-promise rate ~80% on test regions
Wks 2–6	Dwell time-study at Gurgaon Bilaspur + Bhiwandi Mankoli → sets f AND scopes upgrade cost	Hub ops	Measured f with range + capex estimate per hub
Wks 4–10	Dispatch-window + consolidation pilot on the 12 named city corridors	Regional ops (Mumbai, Kolkata, Hyderabad)	Per-corridor breach rate from 69–100% to <50%
Wks 8–12	FTL shift on long-day corridors above half-load	Line-haul planning	Delay ratio on shifted corridors → FTL baseline (≈ 1.9)
Quarterly	Re-rank hubs/corridors; track network excess hours (the un-gameable total), not just the named list	Analytics	Total excess hours/day trending down

Honesty notes, kept short: all breach figures use a constructed SLA (promise = $2 \times$ OSRM, 20% margin; breach rate moves 49%→16% across margins 1.0–1.5 $\times$, so treat the absolute level as a dial and the *rankings* as the finding). Comparisons across route types are observational. Data covers 22 non-peak days; festival-season bottlenecks will be worse and are unmeasured.